

The G12 Phasecheck Interferometer: A Detailed Walkthrough of the Setup, the Locking Protocol, and the Phase Spectrum

Internal explanatory note

Abstract

The G12 “phasecheck” setup is a single-station fibre Mach–Zehnder interferometer (MZI) built around the reflection arm of a SiV nanophotonic cavity. Its purpose is to extract the *frequency-dependent phase* of the cavity’s reflection coefficient $r_c(\omega) = |r_c(\omega)|e^{i\phi_c(\omega)}$, from which the coupling regime (over-coupled, critically coupled, under-coupled) of the cavity is read off directly. This note explains the role of every component on the G12 schematic (`g12Inter_exp_setup.pdf`), derives the APD signal mathematically, distinguishes “locking” from “probing,” explains what $\delta\varphi_{\text{AOM}}$ means, what the “0th” label refers to, and how the free spectral range (FSR) of the unbalanced MZI is measured and compensated. The exposition closely follows the architecture established in Liu *et al.* (Nature 2026, supplementary §4) for the Lukin-group entanglement-assisted telescope, of which this G12 rig is the single-node phase-readout subsystem.

1 Goal of the experiment, in one paragraph

A photon incident on a single-sided Fabry–Perot-like cavity acquires, upon reflection, a complex amplitude $r_c(\omega) = |r_c(\omega)|e^{i\phi_c(\omega)}$ that depends on the input-mirror reflectivity, the internal losses, and the detuning $\omega - \omega_0$ from the cavity resonance ω_0 . The *magnitude* $|r_c|$ alone cannot distinguish over-coupling ($\kappa_{\text{ext}} > \kappa_{\text{int}}$) from under-coupling ($\kappa_{\text{ext}} < \kappa_{\text{int}}$): both give a dip in reflected intensity, identical near critical coupling. The *phase* $\phi_c(\omega)$ does distinguish them: an over-coupled cavity winds the phase by a full 2π across resonance, while an under-coupled cavity makes a small excursion that returns to its baseline.¹ The G12 phasecheck interferometer is built to measure $\phi_c(\omega)$ across ~ 270 GHz of laser detuning around the SiV optical transition, so that this winding—and therefore the coupling regime—can be read off.

2 Architectural map: a fibre Mach–Zehnder with the cavity in one arm

The interferometer is a textbook **unbalanced Mach–Zehnder** (Fig. 1). Two fibre 50:50 (or near-50:50) couplers form the input and recombining beamsplitters of the MZI. The two arms are:

Signal (“upper”) arm. Light goes *up* through the protection tube to the cryostat, through the free-space AOM whose *1st-order* diffracted beam enters the optical circulator, then a 99:1 tap (1% for monitoring, 99% on-cavity), and finally reflects off the SiV nanophotonic cavity. The cavity-reflected light retraces its path back through the 99:1 tap and the circulator. The circulator’s non-reciprocity routes the returning light to a different fibre port than the incoming light, so the AOM is single-passed.

¹Sigg *et al.*, Appl. Opt. **39**, 3638 (2000); see also the discussion of phase response in nanophotonic SiV cavities at Knall *et al.*, Phys. Rev. Lett. **129**, 053603 (2022).

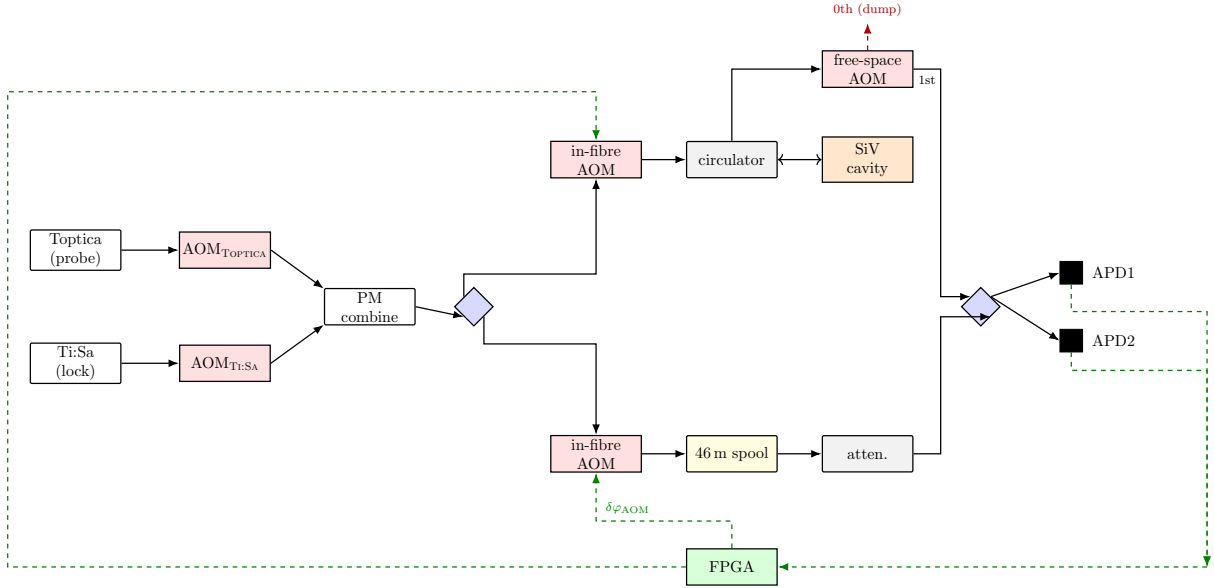


Figure 1: Schematic of the G12 phasecheck interferometer. The input fibre carries both the Ti:Sa lock laser and the Toptica probe laser, interleaved in time via their preparation-side AOMs. A first fibre 50:50 splits the light into the signal arm (free-space AOM, circulator, SiV cavity) and the LO arm (in-fibre AOM, 46 m fibre spool, adjustable attenuator). The arms recombine on a second fibre 50:50, with APD1 and APD2 reading the two complementary outputs. The FPGA uses the APD counts during the lock window to update the common RF phase $\delta\varphi_{\text{AOM}}$ driving both AOMs.

Local oscillator (“lower”) arm. Light goes through the *in-fibre* AOM and a polariser, then through the ~ 46 m fibre spool (a deliberate added delay), then an adjustable attenuator (to match the per-arm power that hits the recombining beamsplitter).

The two arms recombine on a fibre 50:50 and the two complementary output ports are read by avalanche photodiodes APD1 and APD2.

The “laser prep.” box at the bottom of the schematic combines two lasers into the single *Probe + Phaselocking* fibre that feeds the G12 table:

- **g12 “new Ti:Sa”** (the *lock* laser): narrow-linewidth, *frequency-fixed* at $\omega_L/2\pi = 406.699972$ THz. Stable enough to act as the phase reference for the servo loop.
- **g12 Toptica** (the *probe* laser): the same wavelength range but *frequency-tuneable*. This is the laser that is swept to characterise the cavity’s phase response.

After their respective AOM gates and polarisation conditioning, both lasers enter the same polarisation-maintaining (PM) fibre and arrive at the G12 table on a single optical mode.

3 Component-by-component: what each part is doing, and why

3.1 The two lasers, the AOMs in laser prep, and the PM combiner

The two lasers play distinct roles dictated by their stability characteristics. The Ti:Sa is a high-end ring cavity laser locked to a stable reference; its frequency drift over the timescale of a phase-lock servo (μs – ms) is negligible. It is therefore the *phase reference* of the interferometer. The Toptica, on the other hand, is electronically tuneable: its piezo / current can be ramped to

scan ~ 270 GHz, but it is not stable enough to serve as its own reference between sweep points. The Toptica is therefore used solely as the *probe*.

Each laser passes through its own (preparation-side) AOM before being combined into the polarisation-maintaining fibre that feeds the G12 table. The role of the preparation AOM is **purely as a fast optical switch**: depending on a TTL bit from the HDAWG, the AOM RF is either ON (laser light gates through, ~ 1 st-order) or OFF (laser is extinguished, ~ 0 th-order goes nowhere useful). This is exactly the function we see in the HDAWG snippet from the lab report:

```
const newTisa = (1<<30);
...
setDIO(both_aom_on | trigger | newTisa); // newTisa AOM ON
wait(10);
setDIO(both_aom_on);                      // newTisa AOM OFF
```

These prep-side AOMs are the gates that implement the time-interleaving (Sec. 7).

3.2 The two on-table AOMs and what $\delta\varphi_{\text{AOM}}$ represents

On the G12 table there are two more AOMs, and these are the ones that play a role *inside the interferometer*:

Free-space AOM (signal arm, near the cryostat). Diffracts the incoming light. The **1st-order** beam (frequency-shifted by Ω_{RF} , picking up the RF phase $\delta\varphi_{\text{AOM}}$ as an optical phase shift) is what is sent through the circulator and into the SiV cavity. The 0th-order is the residual undiffracted leakage—unwanted “waste light”—and is labelled on the schematic only to make explicit which order is which (see Sec. 3.3).

In-fibre AOM (LO arm). Same physics inside a fibre-pigtailed package. Its 1st-order beam, frequency-shifted and phase-shifted by $\delta\varphi_{\text{AOM}}$, continues through the polariser, the 46 m spool and the attenuator into the recombining 50:50.

What $\delta\varphi_{\text{AOM}}$ means. An AOM is driven by an RF tone $V_{\text{RF}}(t) = V_0 \cos(\Omega_{\text{RF}}t + \delta\varphi_{\text{AOM}})$. The 1st-order diffracted optical beam picks up the frequency shift Ω_{RF} and the instantaneous RF phase $\delta\varphi_{\text{AOM}}$ as an optical phase shift: $E_{1\text{st}}(t) \propto E_{\text{in}}(t) e^{i\Omega_{\text{RF}}t + i\delta\varphi_{\text{AOM}}}$. Thus $\delta\varphi_{\text{AOM}}$ is a *control knob*: a piezo-free, GHz-bandwidth electronic phase shifter for the optical light passing through that arm. Per your clarification, the FPGA outputs a single RF phase $\delta\varphi_{\text{AOM}}$, which is split and fed to *both* on-table AOMs with the same value. With opposite-sign 1st-order assignments in the two arms (a standard trick), the common RF phase translates into a *differential* optical phase across the MZI:

$$\Delta\varphi_{\text{AOM}}^{(\text{diff})} = (+\delta\varphi_{\text{AOM}}) - (-\delta\varphi_{\text{AOM}}) = 2\delta\varphi_{\text{AOM}}, \quad (1)$$

so that changing $\delta\varphi_{\text{AOM}}$ at the FPGA *directly steers the interferometer fringe*.² In what follows we absorb this factor of 2 into the definition of $\delta\varphi_{\text{AOM}}$ for compactness: $\delta\varphi_{\text{AOM}} \equiv 2\delta\varphi_{\text{AOM}}^{(\text{electronic})}$.

Are the AOMs in the feedback loop? Yes. The on-table AOMs are the *actuators* of the lock loop. Their RF phase is the variable the FPGA writes to in order to keep the interferometer parked at a chosen fringe operating point.

²If both arms used the same-sign 1st order, the optical phases would be common-mode and the FPGA would have no leverage. The physical arrangement (which face of the AOM the beam enters, which order is selected) is therefore not cosmetic; it is what makes the loop closeable.

3.3 The “0th” label, demystified

The yellow rectangle near the top of the schematic, marked “0th,” is the *0th-order undiffracted beam* from the free-space AOM in the signal arm. The 1st-order—frequency-shifted and carrying the phase $\delta\varphi_{\text{AOM}}$ —is the one that is sent on to the circulator and the SiV cavity. The 0th-order beam goes nowhere useful (it is typically dumped into a beam block) and serves no functional role in the interferometer. The label is there to make the schematic unambiguous about which order is the working beam. There is no separate “0th” phase-locking branch in this single-station rig (unlike the two-station Lukin Nature 2026 layout where a 0th-order bypass would be more substantive).

3.4 The ~ 46 m fibre spool

The fibre spool is a *deliberate path delay* added to the LO arm to roughly path-match the long signal arm. The signal arm has to leave the fibre breakout, climb the fridge stack, hit the cavity, and come back; this contributes a non-trivial optical length. From the lab report:

$$(920 - 698) \text{ ns} \times \frac{c}{n_{\text{fibre}}} = (220 \text{ ns}) \times \frac{3 \times 10^8}{1.465} \text{ m s}^{-1} \approx 45.46 \text{ m},$$

which is why a ~ 46 m spool was added to the LO arm. The residual mismatch after the spool is what sets the MZI free spectral range; we will return to this in Sec. 8.

Why coarse path-matching matters. If the arm imbalance is too large compared to the laser coherence length, the interferometer visibility collapses. For a laser with coherence time τ_c , fringe visibility falls as $V \propto e^{-(\Delta\tau/\tau_c)^2}$. The Toptica scan range is also relevant: the fringe density scales like $\Delta\tau$, so a long imbalance means rapid fringes, which the FPGA may not be able to track and which makes phase unwrapping harder. The spool is sized to bring the arms to within a few centimetres in residual delay.

3.5 Circulator, 99:1 couplers, Protection Tube

The **circulator** routes incident light into the cavity and the cavity-reflected light to a different output port, separating “go” from “return” on a single fibre. Without it, the back-reflected light from the cavity would have to be picked off with a passive coupler, costing 50% of both incident and returning amplitudes.

The two 99:1 **couplers** along the signal arm are weak taps. They allow real-time monitoring of, for example, the power going into the cavity or the rejected back-reflected power, without significantly attenuating the main beam. They do not change the interferometer physics.

The **Protection Tube** is a mechanical isolation sleeve protecting the fibre as it enters the cryostat / climbs the experimental tower. Mechanically it reduces flexure-induced phase noise (a major source of slow drift in fibre MZIs).

3.6 Polarisers and adjustable attenuators

Fringe visibility is maximised only when the two interfering arms are in the same polarisation state. The in-line polarisers fix the polarisation state on each arm. The adjustable attenuators allow the optical power on the two arms to be matched at the recombining 50:50, since maximum fringe contrast is achieved when $|E_S| = |E_L|$. They also let the operator set the absolute APD count rate (avoiding APD saturation while keeping the shot noise small enough for the lock loop).

3.7 APD1 and APD2

The two avalanche photodiodes detect the two complementary outputs of the recombining 50:50. “Complementary” means: when APD1 sits on a fringe maximum, APD2 sits on a fringe minimum, with a π phase offset. Reading both outputs (rather than one) lets the operator:

1. Form a common-mode-rejected normalised signal $(I_1 - I_2)/(I_1 + I_2) = V \cos(\Delta\varphi)$, removing laser intensity drift,
2. Lock either APD’s counts to the desired fringe operating point with full sign information.

4 The APD signal: mathematical model

4.1 Field amplitudes at the recombining 50:50

Write the optical field arriving at the recombining 50:50 from each arm as

$$E_S(t, \omega) = a_S r_c(\omega) e^{i\omega t} e^{i\varphi_S(\omega)} e^{+i\delta\varphi_{\text{AOM}}}, \quad (2)$$

$$E_L(t, \omega) = a_L e^{i\omega t} e^{i\varphi_L(\omega)} e^{-i\delta\varphi_{\text{AOM}}}, \quad (3)$$

where:

- $r_c(\omega) = |r_c(\omega)|e^{i\phi_c(\omega)}$ is the complex cavity reflection coefficient,
- $\varphi_{S/L}(\omega) = \omega \tau_{S/L}$ are the linear path-length phases of the two arms (drifts slowly),
- the $\pm\delta\varphi_{\text{AOM}}$ are the two AOM optical phases (opposite-sign 1st orders as discussed).

A fibre 50:50 in the convention with a $\pi/2$ phase on the cross-port outputs the two fields

$$E_{\pm} = \frac{1}{\sqrt{2}}(E_S \pm iE_L). \quad (4)$$

The APD photocurrents are proportional to the time-averaged intensities

$$I_1 = |E_+|^2 = \frac{1}{2} [a_S^2 |r_c|^2 + a_L^2] + a_S a_L |r_c| \sin[\Delta\varphi(\omega)], \quad (5)$$

$$I_2 = |E_-|^2 = \frac{1}{2} [a_S^2 |r_c|^2 + a_L^2] - a_S a_L |r_c| \sin[\Delta\varphi(\omega)], \quad (6)$$

with the total interferometer phase

$$\Delta\varphi(\omega) = \underbrace{\omega \Delta\tau}_{\text{linear MZI background}} + \underbrace{\phi_c(\omega)}_{\text{cavity phase}} + \underbrace{2\delta\varphi_{\text{AOM}}}_{\text{FPGA control}} + \underbrace{\varphi_0(t)}_{\text{slow drift}}. \quad (7)$$

Here $\Delta\tau = \tau_S - \tau_L$ is the residual delay imbalance after the 46 m spool. The slow drift term $\varphi_0(t)$ captures thermal and mechanical fluctuations of the fibre paths.

4.2 The contrast/visibility form

It is often more useful to write

$$I_{1,2}(\omega, \delta\varphi_{\text{AOM}}) = \bar{I}(\omega) [1 \pm \mathcal{V}(\omega) \sin(\omega\Delta\tau + \phi_c(\omega) + 2\delta\varphi_{\text{AOM}} + \varphi_0)], \quad (8)$$

with mean intensity $\bar{I} = \frac{1}{2}(a_S^2 |r_c|^2 + a_L^2)$ and visibility

$$\mathcal{V}(\omega) = \frac{2a_S a_L |r_c(\omega)|}{a_S^2 |r_c(\omega)|^2 + a_L^2}, \quad \mathcal{V} = 1 \text{ when } a_S |r_c| = a_L. \quad (9)$$

This is why the adjustable attenuators are present: they let the operator match a_L to $a_S |r_c|$ on average and maximise the contrast.

5 Phase *locking*: what it means, and how it works

What is being “locked.” The interferometer phase $\Delta\varphi$ in Eq. (7) drifts on the millisecond-to-second timescale due to thermal expansion of the fibres, acoustic vibrations of the spool, and slow mechanical motion of the cryostat. “Locking” means: *driving $\delta\varphi_{\text{AOM}}$ in real time so that $\Delta\varphi$ stays at a constant target value.* It does *not* mean stopping the underlying drifts. It means actively cancelling them.

Why mid-fringe. The APD signal in Eq. (8) is sinusoidal in $\Delta\varphi$. Its derivative is $\partial I_1/\partial\Delta\varphi = \bar{I}\mathcal{V}\cos(\Delta\varphi)$, which is maximal in magnitude at $\Delta\varphi = 0, \pi, \dots$ (i.e. at the points where $\sin(\Delta\varphi) = 0$ and the APD sits at the half-fringe point). Locking there makes the response of the APD counts to a small phase perturbation *linear* and *maximally sensitive*:

$$\delta I_1 \approx \bar{I}\mathcal{V}\delta(\Delta\varphi). \quad (10)$$

This is exactly the procedure described in the Nature 2026 supplementary, §4: “We lock the APD photon counts to the reference port count, of which level roughly matches the steepest slope point of the interference pattern.”³

Closing the loop. During the *lock window*, the FPGA:

1. Reads the APD1 photon count rate N_1 during a short integration gate ($\sim\mu\text{s}$ scale).
2. Computes the error $e = N_1 - N_{1,\text{set}}$, where $N_{1,\text{set}}$ is the mid-fringe set point.
3. Updates $\delta\varphi_{\text{AOM}} \leftarrow \delta\varphi_{\text{AOM}} - g e$ (proportional gain g ; typically a PI loop in practice).
4. Writes the new $\delta\varphi_{\text{AOM}}$ to the RF source driving both on-table AOMs.

At each new sample, $\Delta\varphi$ is pushed back towards its mid-fringe target. The user-chosen mid-fringe point is set by an *offset* $\delta\varphi_{\text{AOM}}^{\text{Offset}}$ added on top of the servo correction (in the Nature 2026 paper this is $\delta\varphi_{\text{Offset}}$; in the lab report the analogous `calib.phase` parameter), so that the interferometer can be parked at any chosen $\Delta\varphi$ on the fringe—including, for example, the destructive-interference point used during entanglement generation in Nature 2026.

6 Phase *probing*: how the cavity phase is extracted

The Toptica-driven probe pulse is what actually carries the cavity information. To extract $\phi_c(\omega_{\text{probe}})$ at a chosen probe frequency, the experiment proceeds in two complementary modes:

(a) Phase scan at fixed probe frequency. Hold ω_{probe} fixed and *step $\delta\varphi_{\text{AOM}}$ through a full $[0, 2\pi)$* . Record APD1 (and APD2) counts at each step. The result, $N_1(\delta\varphi_{\text{AOM}})$, is a sinusoid:

$$N_1(\delta\varphi_{\text{AOM}}) = A + B \sin(2\delta\varphi_{\text{AOM}} + \Phi(\omega_{\text{probe}})), \quad \Phi(\omega_{\text{probe}}) = \omega_{\text{probe}}\Delta\tau + \phi_c(\omega_{\text{probe}}) + \varphi_0. \quad (11)$$

A nonlinear least-squares fit to Eq. (11) extracts $\Phi(\omega_{\text{probe}})$. The 2D heatmaps in your lab report—“`apd1`” with vertical axis “freq offset (GHz)” and horizontal axis “phase”—are exactly this: one column for each phase step, repeated at many probe frequencies. The bright/dark diagonal stripes are loci of constant Φ , and their slope encodes $\Delta\tau$ (i.e. the FSR).

³Liu *et al.*, Nature 2026, Suppl. §4.

(b) Quadrature pair. A cheaper measurement (used in Nature 2026, Fig. S10) takes only *two* probe pulses per frequency point: one at $\delta\varphi_{\text{AOM}} = 0$ and one at $\delta\varphi_{\text{AOM}} = \pi/2$ (the latter implemented by toggling the AOM RF phase by 90°). From the in-phase and quadrature APD readouts I_0 and $I_{\pi/2}$,

$$\tan \Phi = \frac{I_0 - \bar{I}}{I_{\pi/2} - \bar{I}} \Rightarrow \Phi(\omega_{\text{probe}}) = \text{atan2}(I_0 - \bar{I}, I_{\pi/2} - \bar{I}). \quad (12)$$

The G12 report uses mode (a) (full phase ramp) because it is more robust against drift between probe points and because it lets the operator inspect the fringe shape directly.

Locking vs. probing in one line. *Locking* steers $\delta\varphi_{\text{AOM}}$ to keep $\Delta\varphi$ at a setpoint; the cavity phase ϕ_c enters only through its (essentially constant) value at the lock laser frequency. *Probing* sweeps the probe frequency and reads out $\Phi(\omega)$; the linear background $\omega\Delta\tau$ is subtracted post hoc (Sec. 8) to recover $\phi_c(\omega)$.

7 The interleaved Ti:Sa / Toptica sequence

The Ti:Sa lock laser and the Toptica probe laser are **interleaved in time**, not run simultaneously. The interleaving uses the preparation-side AOMs (one per laser) as fast gates. The pattern, mirroring the Nature 2026 protocol (Suppl. Fig. S9a) and consistent with the HDAWG `newTisa = (1<<30)` bit toggling in your lab report, is:

Window	State
LOCK ($\sim 50\%$ DC)	Ti:Sa AOM ON , Toptica AOM OFF . APD reads lock fringe; FPGA updates $\delta\varphi_{\text{AOM}}$.
PROBE ($\sim 50\%$ DC)	Ti:Sa AOM OFF , Toptica AOM ON . APD reads probe fringe at ω_{probe} .

There are three reasons for interleaving rather than running both lasers continuously:

1. **Clean APD signal.** If both lasers were on at once, the APD would see the sum of the two fringe patterns, $I_1^{\text{lock}}(\omega_L) + I_1^{\text{probe}}(\omega_{\text{probe}})$, and they would have to be separated either by RF-demodulation (different AOM drive frequencies) or by spectral filtering. Time-interleaving is far simpler.
2. **Avoid disturbing the SiV.** The Ti:Sa lock laser is intentionally placed ~ 21 GHz from the SiV transition (well outside its ~ 100 MHz linewidth) so as not to drive the spin or excite the centre. But even so, having the lock laser on during a sensitive probe window adds shot noise on the same APD; turning it off during probing maximises signal-to-noise. (In the Nature 2026 entanglement protocol, this constraint is even stronger: SiV nuclear spins must be perfectly undisturbed during the entanglement window, which forces strict interleaving.)
3. **Locking bandwidth.** A $\sim 50\%$ lock duty cycle of, say, $10\mu\text{s}$ on / $10\mu\text{s}$ off gives the FPGA a ~ 50 kHz effective update rate—more than enough to cancel thermal/acoustic drifts that live below 1 kHz.

In your data, the parameter `Locklaser_freq_THz`: `406.699972` (Ti:Sa) and the swept `calib_laser_freq_THz` `406.720` (and other Toptica targets) make explicit that the two lasers act at *different* frequencies on the *same* fibre, interleaved by the gating AOMs.

8 Free spectral range, measurement, and compensation

8.1 Definition

Even after path-matching with the 46 m spool, a residual delay imbalance $\Delta\tau$ remains. The linear-in-frequency background $\omega\Delta\tau$ in Eq. (7) produces a sinusoidal fringe in the APD signal as ω is swept. The *free spectral range* is the period of that fringe:

$$\nu_{\text{FSR}} = \frac{1}{\Delta\tau} = \frac{c}{n_{\text{fibre}} \Delta L_{\text{residual}}}. \quad (13)$$

For the values in your report,

$$\nu_{\text{FSR}} = 3.5123 \text{ GHz} \iff \Delta L_{\text{residual}} = \frac{c}{n_{\text{fibre}} \nu_{\text{FSR}}} = \frac{3 \times 10^8}{1.465 \times 3.5123 \times 10^9} \text{ m} \approx 5.83 \text{ cm},$$

which is the few-cm path mismatch left after the spool addition. The independently fitted value $\Delta L_{\text{diff}} = -7.93 \text{ cm} \pm 1.79 \text{ cm}$ in the time-delay measurement is consistent with this.

8.2 Measuring ν_{FSR} : “go out of the cavity”

You want to measure $\Delta\tau$ without the cavity phase $\phi_c(\omega)$ contaminating the result. The procedure in the lab report—“*now, to precisely determine FSR, first go out of the cavity*”—means: tune the probe far enough from the SiV resonance that $\phi_c(\omega)$ is effectively flat across the scan range. Then Eq. (7) reduces to

$$\Phi(\omega) \approx \omega \Delta\tau + \text{const.} \quad (14)$$

A linear fit of $\Phi(\omega)$ vs. ω over a wide scan (e.g. the 30-GHz Toptica ramps shown in your report from 720–750 GHz, 675–705 GHz, etc.) extracts $\Delta\tau$ and hence $\nu_{\text{FSR}} = 1/\Delta\tau$. Your report’s iterated refinements (first fit $\nu_{\text{FSR}} = 3.5390 \text{ GHz}$ with one compensation guess, recompensate with 3.6 GHz, then refit to $\nu_{\text{FSR}} = 3.5123 \text{ GHz}$) are exactly the standard self-consistent iteration: a 1% error in the assumed ν_{FSR} leaves a residual slope in the corrected phase that itself gives the correction to ν_{FSR} .

8.3 Compensating for the FSR background

Once ν_{FSR} is known, the cavity-only phase is obtained by subtracting the linear background:

$$\phi_c(\omega) = \Phi(\omega) - \omega \Delta\tau - \phi_{\text{calib}} = \Phi(\omega) - \frac{\omega}{\nu_{\text{FSR}}} \cdot 2\pi - \phi_{\text{calib}}. \quad (15)$$

The constant ϕ_{calib} is set by the operator’s choice of zero (in your config: `calib_laser_freq.THz: 406.720`, `calib_phase: 0`). The compensation can equally be implemented physically (by adding/subtracting a few centimetres of fibre to drive $\Delta\tau \rightarrow 0$) but the digital subtraction is what is actually done in the analysis pipeline.

8.4 Stitching multiple Toptica ranges

A single Toptica ramp covers only $\sim 30 \text{ GHz}$, but the cavity must be characterised over $\sim 270 \text{ GHz}$. Several ramps—470–500, 510–550, 570–600, 620–650, 668–700, 720–750 GHz of probe offset relative to 406 THz in your report—are stitched together. Each segment is independently FSR-compensated and aligned in phase via the calibration point. Your report’s final plot, with the cavity resonance clearly visible as a phase excursion near 585 GHz above the 406 THz reference and the lock-laser position marked near 700 GHz, is the assembled phase spectrum.

9 Reading the coupling regime from the assembled phase

With $\phi_c(\omega)$ in hand, the coupling regime of the SiV cavity is read off the *winding* of the phase across the resonance:

Over-coupled ($\kappa_{\text{ext}} > \kappa_{\text{int}}$). ϕ_c sweeps by 2π across resonance: a full revolution.

Critically coupled ($\kappa_{\text{ext}} = \kappa_{\text{int}}$). ϕ_c jumps by π at resonance, where the reflection amplitude goes to zero.

Under-coupled ($\kappa_{\text{ext}} < \kappa_{\text{int}}$). ϕ_c makes a finite excursion smaller than 2π and returns to its baseline.

Your final corrected plot, in which the phase near the SiV cavity dashed line drops by $\sim 60^\circ$ before climbing by $\sim 150^\circ$ above it, shows a feature that is consistent with an under- or moderately-coupled cavity (a full 2π winding is not visible across the resolved resonance feature). A definitive verdict needs the local feature to be resolved at finer frequency steps and the noise to be tightened; the present sweep is the diagnostic that tells you *whether* that finer measurement is worth doing and *where* to do it.

10 Questions to Francisca and Ya Cheng

- Is the measurement sequence strictly interleaved — we pulse the Ti:Sa to read the fibre-noise fringe, switch it off, then pulse the Toptica to probe the cavity — or are both lasers continuously on and hitting the APDs simultaneously?
- When we say "phase locking," are the AOMs receiving a real-time closed-loop feedback signal $\delta\varphi_{\text{AOM}}$ that physically cancels fibre vibrations? What feedback mechanism is used, a PID? Is the picture that we monitor the APD counts continuously and update the AOM RF phase to hold the interferometer at the mid-fringe point $\Delta\varphi = 0, \pi, \dots$ where the response to phase perturbations is maximally sensitive?
- What does "0th order" mean exactly, and what is the purpose of the free-space AOM above the circulator? Is it controlled, or is it just a passive switch?
- How are we actually extracting the phase value from the APD counts? During probing, is only the Toptica ON and the TiSa OFF? Are we scanning $\delta\varphi_{\text{AOM}}$ over $[0, 2\pi]$ at fixed ω_{probe} to record a sinusoid, and then stepping ω_{probe} , so that we fit

$$N_1(\delta\varphi_{\text{AOM}}) = A + B \sin(2\delta\varphi_{\text{AOM}} + \Phi(\omega_{\text{probe}})), \quad \Phi(\omega_{\text{probe}}) = \omega_{\text{probe}}\Delta\tau + \phi_c(\omega_{\text{probe}}) + \varphi_0?$$

- To confirm the FSR compensation procedure: we tune the laser far off the cavity resonance so the cavity behaves as a flat mirror, sweep the frequency to measure the residual fibre-length mismatch from the fringe period, and then digitally subtract the resulting linear slope from the cavity scans — is that right?
- Are we using polarisation-maintaining fibres throughout the setup? Why are there so many polarisers in the schematic — do the beamsplitters or other components rotate the polarisation?
- Are APD1 and APD2 just phase-shifted by π relative to each other (i.e. complementary fringes)? Do we read out one APD on its own, or do we use the difference $(I_1 - I_2)$ (or the normalised contrast $(I_1 - I_2)/(I_1 + I_2)$) for better SNR?

11 Summary table of components

Component	Role
g12 new Ti:Sa (Lock)	Stable, frequency-fixed laser at 406.699972 THz. Provides the reference fringe that the FPGA locks against.
g12 Toptica (Probe)	Tuneable laser swept ~ 270 GHz around the SiV transition. Reads out cavity phase $\phi_c(\omega)$.
Prep-side AOMs (one per laser)	Fast TTL-gated optical switches. Implement the time interleaving between lock and probe windows.
First fibre 50:50 (input BS)	Splits the combined laser light into the two MZI arms.
Free-space AOM (signal arm)	Phase actuator for the signal arm; its 1st-order goes through the circulator into the cavity. The label “0th” marks the discarded undiffracted leakage.
Optical circulator	Routes input light into the cavity and reflected light to the recombining beamsplitter, on a single fibre.
99:1 couplers (signal arm)	Weak monitor taps; do not affect the interferometer physics.
SiV nanophotonic cavity	The element being characterised. Imprints $r_c(\omega) = r_c(\omega) e^{i\phi_c(\omega)}$ on the reflected light.
Protection Tube	Mechanical isolation for the fibre entering the cryostat; reduces flexure-induced phase noise.
In-fibre AOM (LO arm)	Phase actuator for the LO arm (receives the same $\delta\varphi_{\text{AOM}}$, opposite-sign 1st order).
Polarisers	Match polarisation between the two arms to maximise fringe contrast.
~ 46 m fibre spool	Coarse path-length match to the signal arm; residual mismatch sets the MZI FSR.
Adjustable attenuator (LO arm)	Power balance for maximum fringe visibility.
Second fibre 50:50 (output BS)	Recombines the two arms; complementary outputs to APD1 and APD2.
APD1, APD2	Read the two complementary fringe outputs.
FPGA	Reads APD counts during the lock window, runs the PI loop, writes $\delta\varphi_{\text{AOM}}$ to the RF source.
$\delta\varphi_{\text{AOM}}$	The RF phase of the AOM drive tone—the FPGA’s single control knob—which translates 1:1 into the differential optical phase across the MZI.

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